Vihaan Mishra

ENVIRONMENTAL RISK | EMERGENCY COMMUNICATIONS | COMMUNITY RESILIENCE

HIGHLIGHTS

01 ACADEMIC

- **AP 5s**: Calculus BC, English Language, Computer Science A
- **College Coursework:** Multivariable Calculus, Foothill College · Research Credit (A), Oberlin College

02 RESEARCH

- PJIR: **Peer-Reviewed Publication** — ClearSight AI
- AGU26: **Accepted for Presentation** — Disaster Resilience & Ham Radio
- Environmental Studies Researcher: **Selected Presenter** — Symposium, King's College, London

03 COMMUNITY IMPACT

- **Founder**, TACER Certification Program
- **Lead, TeenCERT,** Emergency Communications
- **Disaster Service Worker + FCC Technician (KO6FXI)**

RESEARCH & PUBLICATIONS

1. Environmental Researcher, Authored Original Research Paper

Peer-Reviewed Publication: Mishra, V. (2026). "HybridEnhancementNet: A Novel Deep Learning Architecture for Low-Light Image Enhancement through Model Fusion." Princeton Journal of Interdisciplinary Research, 2(1), July 2026. [**Published Article]** (ClearSight AI)

Mentor: Dr. Robail Yasrab, MRC Biostatistics Unit, **University of Cambridge, UK**

Presented: Sigma Xi International Forum on Research Excellence (IFoRE '25), Oct. 30–Nov. 1, 2025

Presented: Student Research Symposium, VES Institute of Technology, Mumbai, Jan 7-8, 2026

Key Findings: Achieved 22.51 dB PSNR / 0.791 SSIM, outperforming CIDNet and Zero-DCE; generalized from 500 paired training/test images to 6,000 previously unseen real-world low-light images.

Implementation: The model is being integrated into VES's drone platform for disaster recovery.

2. Emergency Communications Researcher, Authored Original Research Paper

Accepted Conference Presentation: Mishra, V., & Sunter, D. (2026). "Natural Disaster Risk and Ham Radio Backup Capacity in the United States." AGU26 Annual Meeting, American Geophysical Union, San Francisco, CA, Dec. 2026. [**AGU Program Record**]

Mentor: Dr. Deborah Sunter, Civil & Environmental Engineering, **Tufts University, Medford, MA**

Presented: Los Altos Hills HAM Club, CA, Oct. 2, 2026

Impact: Analyzed 3,109 counties using 826,751 active FCC licenses + FEMA and Census data; identified 189 high-risk / low-radio-capacity counties, including 48 also low-income.

Key Findings: Nationwide risk alone was not predictive; hazard-specific analysis identified 252 high-hurricane-risk / low-radio-capacity counties, creating a targeted map for preparedness and radio-training efforts.

3. Community Resilience Researcher, Authored Original Research Paper

Selected Conference Presentation: Mishra, V. (2026). "Examining Wildfire Model Generalizability Through Cross-Region Transfer and Physically Coherent Counterfactual Analysis." Student Research Symposium, King's College, London, UK, July 30–31, 2026. [**Credential**]

Mentor: Dr. Robert Edwin Rouse, **University of Cambridge, UK**

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Presented: FireWise of Los Altos Hills, CA, Oct. 2026

Impact: Achieved similar within-region performance in California (F1 0.687; AUC 0.904) and Spain (F1 0.681; AUC 0.903), but cross-region transfer fell sharply to F1 0.50 (CA→Spain) and 0.18 (Spain→CA)—demonstrating that strong local wildfire models may not generalize across regions.

Key Findings: On 504 observed Spain fire events from 2026, the Spain model identified 375 (74.4%); physically coherent 1°C cooling reduced model fire-positive classifications by 601 in California (−9.5%) and 185 in Spain (−5.2%).

COMMUNITY RESILIENCE LEADERSHIP

Founder & Executive Director, TACER — Teens Assisting with Communications in Emergency Response [https://TACERLeaders.org]

Partnered with Santa Clara County Fire District

- Built TACER Certification into a prerequisite for Santa Clara County TeenCERT, moving students from emergency-communications training to Disaster Service Worker certification and community response leadership
- Designed and expanded a 7-module curriculum for teen (high school and middle school) audiences
- Net Control Operator (NCO), led the first-ever Youth HAM Network in Santa Clara County - Oct 2026

Head of Emergency Communications & Board Member, Santa Clara County Teen CERT

Certified Disaster Service Worker | FCC Technician-class amateur radio operator (KO6FXI)

- Leads emergency-communications work within a countywide teen preparedness network across Santa Clara County
- Connects youth training, radio readiness, and disaster-response drills for emergency response.

Civic & School Leadership

Vice-Chair, Los Altos Hills Youth Commission | Treasurer, Los Altos High School ASB

- Invited to present disaster-risk and ham radio research to a local amateur-radio audience
- Convener, Youth Leadership Conference on AI and Cybersecurity

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ACADEMIC ACHIEVEMENT

- **University of Rochester Xerox Award for Innovation & Information Technology —** School Faculty-selected junior; Rochester permits one nominee per participating school for this award.
- **MVLA / Los Altos High School Student of the Month —** District recognition for community service, research, technology, and leadership.

METHODS

Community resilience · Natural hazards · Emergency communications · Geospatial analysis · Remote sensing · Counterfactual analysis · Statistical analysis · Scientific communication · Machine Learning